

Synthetic Dataset Construction Pipeline

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1 Pipeline Core Design

The benchmark is built from hidden structure before any retrieval text exists. This is the important reversal from the old MIMIC-IV-only workflow: instead of starting from raw clinical notes and hoping a clean coverage problem appears, the pipeline first defines the multi-facet evidence structure, then renders it into clinical prose, embeds it, filters it by geometry, and finally evaluates retrieval.

1. define a controlled ontology of conditions, subgroup labels, and two clinical axes;
2. enumerate hidden query plans with four required answer facets;
3. optionally calibrate which facet should be overrepresented using probe embeddings;
4. generate structured clinical facts for gold facets, near-miss distractors, and background outlier clusters;
5. render those facts into validated clinical-style chunks;
6. render natural-language queries and canonical gold answers;

7. derive qrels directly from the hidden chunk–facet mapping;
8. embed queries and chunks, filter queries by coverage-sensitive embedding geometry, and evaluate top- k , MMR, and facility-location.

1.1 Current Benchmark Shape

The reduced benchmark is still deliberately small, but it already has the full design pattern needed for the thesis:

Stage	Design role	Result
Query planning	Define the hidden retrieval problem	Condition, subgroup pair, facet plan, and dominant facet choice
Fact generation	Expand each facet into evidence and distractors	Hidden structured clinical facts with gold and non-gold roles
Chunk rendering	Turn facts into retrievable text	Clinical-style evidence chunks plus query-specific memberships
Query and answer generation	Expose the user-facing task	Natural-language queries and canonical answers
Qrel projection	Fix relevance before retrieval	Query–chunk labels derived from the hidden construction
Embedding and filtering	Check whether the benchmark geometry is usable	Candidate pools where top- k is likely to collapse and coverage can matter
Retrieval evaluation	Compare ranking strategies	Top- k , MMR, and facility-location under the same candidate pool
Diagnostics	Make the benchmark behavior inspectable	Geometry plots and retrieval comparison plots

The cleanest setting for understanding the benchmark is a *query-local* candidate pool. In that setting, each query retrieves from its own controlled local universe: gold chunks, hard near-miss distractors, and optional background outlier clusters. Broader candidate pools can also be studied, but the query-local setting makes the intended mechanism easiest to diagnose.

The current reduced design has these constraints:

- each query has exactly four answer facets: two compared subgroups crossed with `treatment_duration` and `rehab_outcome`;
- the main query families are subgroup comparison and outcome synthesis;
- one facet is intentionally dominant, receiving more evidence than the other three facets;
- hard distractors cycle through same-condition wrong-subgroup, same-subgroup wrong-condition, and same-axis wrong-condition patterns;
- optional background outlier clusters create compact but irrelevant clinical islands inside the query-local pool;
- chunk text can be directly LLM-generated, deterministically templated, or templated and then lightly rewritten;
- queries come from fixed templates, with optional paraphrasing guarded by entity and axis checks;
- gold answers are deterministic summaries of the hidden fact rows, not LLM-authored answers.

The reduced design does not rely on a Prolog solver, agentic retrieval, information-not-found cases, conflicting-note resolution, or LLM-as-judge qrel creation. Those remain possible extensions, but they are not part of the present benchmark story.

1.2 Literature overview

The literature does not contain one paper that already solves this exact thesis problem. The useful part is more practical: several papers show how to build synthetic or semi-synthetic RAG benchmarks where questions, evidence, answers, and labels are controlled instead of guessed after retrieval. The current pipeline takes that construction discipline and applies it to a narrower question: can a coverage method retrieve more complete multi-facet clinical evidence than top- k or MMR?

Most RAG benchmark-generation papers start from an existing corpus and then generate questions or labels on top of it. This pipeline deliberately reverses that order. It first creates the hidden retrieval problem, then renders clinical-style text. This is the main lesson from the failed MIMIC-IV-only attempt: real discharge chunks are clinically rich, but they did not reliably contain the clean facet clusters needed to test facility-location coverage.

1.2.1 Ideas used directly in the implemented pipeline

- **Build the evidence plan before the prose.** MHTS constructs multi-hop questions from planned claims, evidence clusters, and difficulty signals [9]. DRAGON keeps explicit links between documents, queries, clues, answers, and citations [13]. In this pipeline, the same idea becomes a simple hidden artifact chain: a query plan defines the four facets, a fact table expands them into evidence rows, a membership mapping records which chunk supports which facet, and qrels are projected from that mapping. The LLM may write chunk text, but it does not decide the gold labels.
- **Treat a broad query as several answer facets.** ClaimSpect and MQ4CS are useful because they make the sub-aspects of a broad information need explicit [6, 1]. The implemented version is intentionally concrete: every query compares two subgroups across two axes, `treatment_duration` and `rehab_outcome`. This gives exactly four required facets, so retrieval can be evaluated on whether it covers all four, not only whether it retrieves any relevant chunk.
- **Generate the corpus around the question, not independently from it.** EnterpriseRAG-Bench is important because it treats synthetic RAG data as question-aware: documents are generated so that the benchmark has planned evidence, noise, answer facts, and conformance checks [14]. The current reduced implementation uses the part that matters for the thesis: for each query, it creates a local universe with redundant gold chunks, complementary gold facets, near-miss distractors, optional background outlier clusters, and later geometry checks.
- **Use a hidden world, but keep the machinery simple.** PhantomWiki generates a fresh hidden symbolic world and derives questions and answers from it [4]. This pipeline copies the separation between hidden truth and surface text, not the full solver setup. The hidden world here is a set of simple structured tables and logical forms: condition, subgroup, axis, value bin, facet ID, cluster role, and gold/distractor role. This is enough to regenerate qrels and answers deterministically.
- **Control difficulty through retrieval structure.** MHTS argues that difficulty is not only the number of hops; it also depends on how evidence is distributed [9]. Here difficulty is controlled by the number of facets, the number of dominant and complementary gold chunks, the number and type of distractors, the candidate-pool scope, and whether nearest-neighbor top- k collapses onto the dominant facet. That is why `calibrate_plans` and `geom_filter` are part of the benchmark construction rather than optional plotting code.
- **Validate the generated benchmark by its retrieval behavior.** Synthetic test-collection work and RAGVAL show that automatic benchmark construction needs validation, not just fluent generated questions [11, 7]. AGORABench makes the same practical point from another angle: a model’s usefulness as a data generator has to be measured downstream [8]. In this pipeline, a generated query is kept only if the candidate pool has the intended geometry: all facets are visible, top- k is overly concentrated, gold facets are more internally similar than cross-facet pairs, and hard distractors are present.
- **Use diversity carefully.** DATAGEN and the diverse private RAG dataset framework

motivate attribute-guided generation, diversity checks, and cluster-aware sampling [5, 3]. For this thesis, diversity is not just “make examples different”. The benchmark needs planned redundancy inside one dominant facet, separation between required facets, and distractors that are similar enough to be tempting. Removing all near-duplicates would remove the top- k failure case that the experiment is meant to test.

1.2.2 Clinical grounding used in the current pipeline

The medical papers are used for scaffolding, not to claim that the generated chunks are faithful synthetic EHR records. The benchmark is a clinical-style retrieval stress test.

- **Care-path structure.** HealthBranches motivates generating clinical QA data from explicit decision pathways [2]. The current ontology uses a smaller version of that idea: each condition has plausible treatments, treatment-duration bins, and rehabilitation-outcome bins.
- **Cohort-like constraints.** Work on generating patient cohorts from EHRs decomposes clinical criteria into condition, demographic, comorbidity, intervention, and outcome constraints [16]. The implemented query plans use the same style of decomposition: condition plus two subgroup labels plus two clinical axes.
- **Controlled terminology.** Knowledge-infused medical QA systems such as KI-MAG motivate using normalized medical terms and relations to reduce drift in generated QA data [15]. In this pipeline, that becomes a small controlled vocabulary for conditions, subgroup labels, treatments, and outcomes, so chunks do not randomly rename the same clinical concept.
- **Quality reporting for synthetic medical data.** Biomedical synthetic-data reviews emphasize utility, consistency, subgroup balance, and transparent validation [12, 10]. The current pipeline reflects this through deterministic seeds, artifact tables, generation rejects, qrel provenance, split fields, and geometry diagnostics.

MIMIC-IV remains useful as a reference source for condition families, common comorbidities, demographic bins, and discharge-note style. It should not be described as the source of gold evidence in the current pipeline. Gold relevance comes from the hidden construction, not from post-hoc annotation of MIMIC-IV notes.

1.2.3 What is deliberately not part of the reduced implementation

Some ideas in the broader literature are useful later but are not part of the reduced design. The benchmark does not rely on a Prolog solver, an agentic multi-step data-generation system, LLM-as-judge qrel creation, information-not-found questions, conflicting-note resolution, or automatic prompt evolution. Those additions may be interesting, but they would make the advisor story harder to evaluate right now. The present claim is narrower and cleaner: construct a controlled multi-facet clinical retrieval problem, verify that its embedding geometry exposes a coverage failure for top- k , and compare top- k , MMR, and facility-location on facet-level retrieval metrics.

2 Pipeline

2.1 Stage 1: Ontology and Design Parameters

The pipeline starts from a controlled clinical ontology and a small set of benchmark design parameters. The ontology has three blocks:

- **conditions:** display names, related terms, plausible treatments, treatment-duration bins, and rehabilitation-outcome bins;
- **subgroups:** demographic or comorbidity groups, with canonical labels and aliases;
- **clinical_axes:** exactly `treatment_duration` and `rehab_outcome`.

A representative condition sketch is:

```

encephalitis_myelitis:
  display: encephalitis or myelitis
  terms: [encephalitis, myelitis, neurologic deficit, acyclovir]
  treatments: [acyclovir, corticosteroids, supportive care]
  duration_treatments: [acyclovir, corticosteroids]
  duration_days:
    short: [5, 7]
    standard: [10, 14]
    prolonged: [21, 28]
  rehab_outcomes:
    home_rehab:
      - discharged home with outpatient neurorehabilitation
    inpatient_rehab:
      - transferred to inpatient rehabilitation for persistent weakness
  persistent_deficit:
    - persistent focal deficits requiring close neurologic follow-up

```

The main design parameters control:

- how many conditions and subgroup pairs are instantiated;
- which query families are used;
- how much evidence is allocated to the dominant facet versus the complementary facets;
- how many hard distractors and background outliers are added;
- whether chunk text is directly generated, templated, or templated and then rewritten;
- which embedding model is used;
- which candidate-pool scope and retrieval hyperparameters are evaluated;
- which thresholds define a geometry pass.

2.2 Stage 2: Hidden Query Plans

Before any prose is generated, the benchmark creates hidden query plans. Each plan chooses one condition, one unordered subgroup pair, and one query family.

Because there are two subgroups and two axes, every plan expands into four required facets:

1. subgroup A + treatment_duration;
2. subgroup A + rehab_outcome;
3. subgroup B + treatment_duration;
4. subgroup B + rehab_outcome.

Each plan records the condition, subgroup pair, query family, dominant facet, target evidence counts, and split assignment. It also stores a lightweight logical form that makes the hidden structure mechanically checkable.

Example plan sketch:

```

query family: subgroup comparison
condition: encephalitis or myelitis
subgroup A: older than 75
subgroup B: uncomplicated diabetes
dominant facet: subgroup A + treatment_duration
number of facets: 4
facets:
  - subgroup: older than 75
    axis: treatment_duration
    value_bin: short
    cluster_role: dominant_gold
    target_gold_chunks: 18

```

```
- subgroup: older than 75
  axis: rehab_outcome
  cluster_role: complementary_gold
  target_gold_chunks: 8
```

The split assignment is deterministic from the plan index, and surface wording variants are assigned in a stable round-robin way. This keeps the train/validation/test boundary independent from later text generation.

2.3 Stage 3: Dominance Calibration

To make the retrieval problem coverage-sensitive, one facet is intentionally overrepresented. Two modes are useful:

- **Rotating dominance:** alternate the dominant facet by design;
- **Embedding-calibrated dominance:** keep the query wording neutral, render neutral probe texts for each facet, and choose the facet that is naturally closest to the query in embedding space.

This stage matters because it keeps the natural query wording neutral. The dominant facet is not exposed through an obvious lexical anchor; it is chosen so that ordinary nearest-neighbor retrieval is likely to collapse onto one facet even when the query itself looks balanced.

2.4 Stage 4: Structured Clinical Facts

Each query plan expands into hidden clinical facts. These facts are the source of truth for chunks, qrels, and canonical answers.

Gold facts:

- support exactly one facet;
- belong either to the dominant gold cluster or to one of the complementary gold clusters;
- carry the clinical fields needed to render valid evidence later.

Hard distractors:

- are non-gold by construction;
- are designed to confuse a particular facet;
- mainly follow same-condition wrong-subgroup, same-subgroup wrong-condition, and same-axis wrong-condition patterns.

Background outliers:

- are non-gold and not attached to any answer facet;
- form compact but irrelevant clinical islands in the candidate pool.

Axis-specific constraints keep duration evidence and rehabilitation evidence cleanly separable. Structurally equivalent facts can also reuse the same surface chunk across different queries. This is useful because the benchmark needs controlled redundancy, not accidental one-off paraphrases everywhere.

2.5 Stage 5: Chunk Rendering and Normalization

There are three rendering modes:

1. **Direct LLM generation:** the chunk is written under strict evidence constraints;
2. **Deterministic templates:** the chunk is rendered from fixed clinical templates;
3. **Template rewrite:** a deterministic draft is rewritten more naturally, but only kept if it still preserves the required evidence.

Every chunk is validated against required evidence constraints. A kept chunk must mention the condition, the relevant subgroup, and the axis-specific evidence that makes it useful. If a chunk fails those checks, the whole query is rejected, because partial local evidence would weaken the benchmark.

Example rendered chunk:

```
The 82-year-old woman older than 75 was admitted with encephalitis or myelitis and new gait instability. For total treatment duration, acyclovir was continued for 21 days. Neurologic symptoms improved slowly, and the longer course was documented because deficits persisted near discharge.
```

The rendered corpus is separated into two conceptual layers:

- a **document layer**, which is the actual retrieval corpus;
- a **membership layer**, which preserves the query-specific link from each chunk to facts, facets, and cluster roles.

This split matters. Retrieval embeds unique chunk documents, while qrels and facet coverage are query-local through the membership table.

2.6 Stage 6: Queries and Canonical Answers

Queries are rendered from fixed families rather than written freely from scratch. The main query families are:

- **subgroup_comparison**: asks how treatment duration and rehabilitation outcome differ between two subgroups;
- **outcome_synthesis**: asks for therapy-course length and discharge rehabilitation status for subgroup A versus subgroup B.

Example query:

```
For patients diagnosed with encephalitis or myelitis, how do treatment duration and rehabilitation outcome differ between patients older than 75 and patients with uncomplicated diabetes?
```

Optional paraphrasing is allowed only if the condition, both subgroup labels, and both axes remain present. Otherwise the original phrasing is kept.

The canonical answer is deterministic. Duration summaries report the dominant duration pattern and common treatment; rehabilitation summaries report the dominant rehab pattern. Every answer is tied back to facet-level supporting evidence.

2.7 Stage 7: Qrels from Hidden Memberships

Relevance is projected directly from the hidden construction. If a chunk belongs to a gold facet for the query, it is relevant; otherwise it is non-relevant. Hard negatives and background outliers remain distinguishable inside the non-gold set.

This stage is fundamental to the benchmark design. Relevance is not inferred from generated prose, lexical overlap, embedding similarity, LLM judgement, or human annotation. It is projected from the hidden construction.

2.8 Stage 8: Embeddings

Queries and chunk documents are embedded with the same model family. Retrieval uses cosine similarity between the query vector and chunk vectors, $\text{sim}(\text{query}, \text{chunk}) = \mathbf{e}_{\text{query}} \cdot \mathbf{e}_{\text{chunk}}$.

The same embedding space is also used by dominance calibration and geometry filtering. This keeps the retrieval problem honest: the failure mode is created in the same space where retrieval will later be measured.

2.9 Stage 9: Geometry Filtering

A generated query is useful only if its candidate pool actually expresses the intended retrieval geometry. The benchmark therefore filters queries before reporting retrieval results.

Three candidate-pool settings are meaningful:

- **Query-local**: retrieve only from the local evidence universe designed for that query;
- **Same-condition**: retrieve from all chunks associated with the same condition family;
- **Full-corpus**: retrieve from the entire generated corpus.

A query passes when:

- all four gold facets are visible in the candidate pool;
- plain nearest-neighbor top- k over-selects the planned dominant facet;
- within-facet similarity exceeds cross-facet similarity by a sufficient margin;
- enough near-miss distractors are present;
- background outlier clusters are present and complete when they are part of the design.

This stage is what turns synthetic generation into a controlled retrieval benchmark rather than just a synthetic text corpus.

2.10 Stage 10: Retrieval Evaluation

The benchmark compares:

- **top- k** : the nearest-neighbor baseline;
- **MMR**: a dispersion method that rewards novelty relative to already selected chunks;
- **facility-location**: the coverage method of interest.

All methods operate inside the same semantic top- N candidate pool. The facility-location selector combines direct query relevance with marginal coverage gain, $\text{score}(j) = \text{lambda} * \text{sim}(q, j) + (1 - \text{lambda}) * \text{marginal_coverage_gain}(j)$.

The expected thesis behavior is:

- top- k may retrieve many relevant chunks but repeatedly from the dominant facet;
- MMR may reduce redundancy but sometimes spends budget on semantically distant distractors;
- facility-location should prefer representatives of uncovered local clusters when the generated geometry is successful.

2.11 Stage 11: Diagnostics and Visualization

The benchmark produces two kinds of diagnostics:

- **Geometry diagnostics**: visualize candidate pools, hidden facet structure, distractor clusters, and the selections made by top- k , MMR, and facility-location;
- **Evaluation diagnostics**: summarize metric trade-offs across k and λ values.

These diagnostics matter because the thesis claim is geometric. They make it possible to show not only that one method scores higher, but why it scores higher.

3 Data splits

Splits are assigned deterministically from the plan index:

- **train**: default split for most generated plans;
- **validation**: used to inspect generation quality and choose retrieval hyperparameters such as MMR/facility-location λ values;
- **test**: final held-out evaluation.

All benchmark objects inherit the split assigned at planning time, and final reporting should be done on the subset of queries that survive generation and geometry filtering.

4 Prompt and Template Families

Stage	Prompt/template family	Current role
Query plans	Query template families	No LLM. Template variants are assigned in a stable way so surface wording does not drift randomly across the benchmark.
Clinical facts	Structured ontology sampling	No LLM. Facts are generated from conditions, subgroup labels, and axis-specific values under controlled sampling rules.
Chunks	Clinical templates or constrained LLM prompting	Direct generation, deterministic rendering, and rewrite modes are all allowed, but every chunk must preserve the required evidence.
Queries	Query templates with optional paraphrase	Deterministic templates are primary. Paraphrases are accepted only if the condition, both subgroups, and both axes remain explicit.
Gold answers	Deterministic aggregation	No LLM. Answers are synthesized from grouped gold facts and tied back to facet-level supporting evidence.
Qrels	Hidden membership projection	No LLM. Relevance labels come directly from the hidden construction rather than post-hoc annotation.

5 Metrics

Metric	Purpose
FacetCoverage@k	Fraction of hidden answer facets represented by at least one selected gold chunk. This is the main thesis metric.
MeanFacetRecall@k	Mean per-facet recall. This is the main depth-of-coverage summary.
Precision@k	Fraction of selected chunks that are gold for any answer facet.
Recall@k	Fraction of all gold chunks retrieved, regardless of facet. Useful but secondary because the dataset intentionally contains many redundant gold chunks.
MAP@k	Conventional average precision over binary gold labels. Secondary for the same reason as recall.
FacetMRR@k	Mean reciprocal rank of the first selected evidence chunk for each facet.
alpha-nDCG@k	Diversified ranking score that treats facets as subtopics and reduces gain for repeated evidence from the same facet.
AnswerROUGE2Recall@k	Lexical recall between selected chunk text and the canonical answer after query-term filtering.
DistractorRate	Fraction of selected chunks that are non-gold. Lower is better.
NearMissDistractorRate	Fraction of selected chunks from facet-like hard negatives.
BackgroundOutlierRate	Fraction of selected chunks from coherent non-gold background islands.
DominantFacetRate	Fraction of selected chunks whose facet equals the planned dominant facet. High values indicate collapse onto the overrepresented facet.
RedundantGoldRate	Fraction of selected chunks that are gold but do not add a new facet.
Facility-location score	Coverage score of the selected set in embedding space. Diagnostic, not a qrel metric.
Average cosine similarity	Mean query similarity of the selected chunks.
Jaccard overlap with top- k	Set overlap between a method’s selection and the top- k selection.

6 Benchmark Objects

Object	Purpose
Hidden query plans	One record per benchmark question, containing the condition, subgroup pair, query family, four facets, dominant facet, split, and target evidence counts.
Dominance calibration audit	Optional record of why a particular facet was chosen as dominant.
Structured fact table	Hidden source of truth for the benchmark: clinical values, facet support, distractor roles, and validation constraints.
Retrieval corpus	The set of unique chunk documents seen by the retriever.
Membership mapping	Query-specific links from chunks to facts, facets, and cluster roles.
Rejection log	A record of generated items that failed evidence validation and were excluded.
Query set	The natural-language benchmark questions exposed to retrieval.
Canonical answer set	Deterministic answers with facet-level summaries and supporting evidence links.
Qrels	Query–chunk relevance labels projected from the hidden construction.
Embedding store	Vector representations for queries and chunks, plus the metadata needed to reproduce retrieval scores.
Geometry diagnostics	Per-query evidence that the candidate pool has the intended coverage-sensitive structure.
Evaluation results	Per-query retrieval metrics for each method and hyperparameter setting.
Evaluation summaries	Aggregated metrics used for method comparison.
Visualization diagnostics	Plot-oriented records for geometry and selection analysis.

References

- [1] Zahra Abbasiantaeb, Simon Lupart, and Mohammad Aliannejadi. **Generating multi-aspect queries for conversational search**. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics*, pages 8201–8217, 2026. doi: 10.18653/v1/2026.eacl-long.383. URL <https://aclanthology.org/2026.eacl-long.383/>.
- [2] Cristian Cosentino, Annamaria Defilippo, Marco Dossena, Christopher Irwin, Sara Joubbi, and Pietro Liò. **Healthbranches: Synthesizing clinically-grounded question answering datasets via decision pathways**, 2025.
- [3] Ilias Driouich, Hongliu Cao, and Eoin Thomas. **Diverse and private synthetic datasets generation for RAG evaluation: A multi-agent framework**, 2025. URL <https://arxiv.org/abs/2508.18929>.
- [4] Albert Gong, Kamilė Stankevičiūtė, Chao Wan, Anmol Kabra, Raphael Thesmar, Johann Lee, Julius Klenke, Carla P. Gomes, and Kilian Q. Weinberger. **PhantomWiki: On-demand datasets for reasoning and retrieval evaluation**. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025. URL <https://arxiv.org/abs/2502.20377>.
- [5] Yue Huang, Siyuan Wu, Chujie Gao, Dongping Chen, Qihui Zhang, Yao Wan, Tianyi Zhou, Chaowei Xiao, Jianfeng Gao, Lichao Sun, and Xiangliang Zhang. **DATAGEN: Unified synthetic dataset via large language models**. In *International Conference on Learning Representations*, 2025.
- [6] Priyanka Kargupta, Runchu Tian, and Jiawei Han. **Beyond true or false: Retrieval-augmented hierarchical analysis of nuanced claims**. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, pages 29664–29679, 2025. doi: 10.18653/v1/2025.acl-long.1434. URL <https://aclanthology.org/2025.acl-long.1434/>.
- [7] Tristan Kenneweg, Philip Kenneweg, and Barbara Hammer. **RAGVAL: Automatic dataset creation and evaluation for RAG systems**. In *2024 2nd International Conference on Foundation and Large Language Models*, pages 470–475, 2024. doi: 10.1109/FLLM63129.2024.10852482.
- [8] Seungone Kim, Juyoung Suk, Xiang Yue, Vijay Viswanathan, Seongyun Lee, Yizhong Wang, Kiril Gashtevski, Carolin Lawrence, Sean Welleck, and Graham Neubig. **Evaluating language models as synthetic data generators**. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, pages 6385–6403, 2025. URL <https://aclanthology.org/2025.acl-long.320/>.
- [9] Jeongsoo Lee, Daeyong Kwon, Kyohoon Jin, Junnyeong Jeong, Minwoo Sim, and Minwoo Kim. **MHTS: Multi-hop tree structure framework for generating difficulty-controllable QA datasets for RAG evaluation**, 2025. URL <https://arxiv.org/abs/2504.08756>.
- [10] Larissa Montenegro, Luis M. Gomes, and José M. Machado. **What we know about the role of large language models for medical synthetic dataset generation**. *AI*, 6(6):109, 2025. doi: 10.3390/ai6060109.
- [11] Hossein A. Rahmani, Nick Craswell, Emine Yilmaz, Bhaskar Mitra, and Daniel Campos. **Synthetic test collections for retrieval evaluation**. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2647–2651, 2024. doi: 10.1145/3626772.3657942. URL <https://dl.acm.org/doi/10.1145/3626772.3657942>.
- [12] Hanshu Rao, Weisi Liu, Haohan Wang, I-Chan Huang, Zhe He, and Xiaolei Huang. **A scoping review of synthetic data generation by language models in biomedical research and application: Data utility and quality perspectives**. *Journal of Healthcare Informatics Research*, 10:367–392, 2026. doi: 10.1007/s41666-026-00229-9.
- [13] Haiyang Shen, Hang Yan, Zhongshi Xing, Mugeng Liu, Yue Li, Zhiyang Chen, Yuxiang Wang, Jiuzheng Wang, and Yun Ma. **DRAGON: Domain-specific robust automatic data generation for RAG optimization**. In *Findings of the Association for Computational Linguistics: EACL*, pages 1065–1078, 2026. doi: 10.18653/v1/2026.findings-eacl.56. URL <https://aclanthology.org/2026.findings-eacl.56/>.
- [14] Yuhong Sun, Joachim Rahmfeld, Chris Weaver, Weijia Chen, Roshan Desai, Wenxi Huang, and Mark H. Butler. **EnterpriseRAG-Bench: A RAG benchmark for company internal knowledge**, 2026. URL <https://arxiv.org/abs/2605.05253>.
- [15] Zafar et al. **KI-MAG: A knowledge-infused abstractive question answering system in medical domain**, 2024.
- [16] Angelo Ziletti and Leonardo D’Ambrosi. **Generating patient cohorts from electronic health records using two-step retrieval-augmented text-to-SQL generation**, 2025.